



RESTORE4LIFE — DANUBE WETLANDS HYDROAPP

Connecting your Earth Engine account

From a bare Google account to a working connection: creating the Cloud project, registering it for noncommercial use, and handing it to the application.

<https://restore4life.icts-donana.es>

Why an account of your own

Everything this application computes — hydroperiods, anomalies, terrain wetness, statistics, exports — runs on Google Earth Engine. It does not run on a shared institutional account: each user connects their own, and every computation is charged against their own quota and runs under their own credentials.

That has three consequences worth knowing before you start:

- You need a Google Cloud project registered for Earth Engine. Getting one is free for the research and conservation work this project does, and the whole of this manual is about obtaining it.
- Exports go to **your** Google Drive, and long-running export tasks are followed from **your** Earth Engine task manager.
- When a computation is refused for being too large, it is your project's monthly allowance that is the limit — see [how much computation you get](#).

The application stores the token that lets it act on your behalf, encrypted, and nothing else about your Google account. You can withdraw it at any moment from the same screen that granted it.

Set aside about fifteen minutes for the whole of this the first time. Once done, it is done: the connection survives signing out and is only made again if you disconnect it or Google withdraws it.

Step 1 — A Google account

Any Google account works, whether it is a personal `@gmail.com` or an institutional one. If your institution runs its accounts through Google — as the CSIC does — use the institutional one: eligibility for free noncommercial access is judged on the organisation behind the account, and an institutional address makes that case for itself.

One caution, and it is the source of most of the confusion later on: if you are signed in to several Google accounts in the same browser, every step from here on can silently land on the wrong one. Before starting, either sign out of all but one, or do the whole procedure in a separate browser profile.

Step 2 — The Cloud project, registered

Earth Engine works through a Google Cloud project. The project is a container and a name; you are not buying anything by creating one, and no card is asked for on the noncommercial path.

Go to console.cloud.google.com/earth-engine (<https://console.cloud.google.com/earth-engine>). This page creates the project and registers it in one pass, which is why it is worth starting here rather than creating a

project by hand and registering it afterwards.

 **Screenshot pending**

`static/img/manual/cloud-onboarding.png`

The Earth Engine page of the Google Cloud console, before any project exists.

The Earth Engine setup page of the Cloud console. It creates the project and registers it in one pass, which is why it is worth starting here rather than making the project by hand.

- 1 The project this page is about. Empty on a first visit — the next step is what fills it in.
- 2 Starts the questionnaire that decides whether the project is commercial or noncommercial.

If you have no project yet, create one. Give it a name you will recognise in six months — `restore4life-yourname` reads better than `my-project-84321` — and accept the identifier Google proposes.

Then work through the registration questionnaire. It decides whether the project is commercial or noncommercial, and the answer for everything done with this application is **noncommercial**.

 **Screenshot pending**

`static/img/manual/register-noncommercial.png`

The registration questionnaire with the noncommercial option selected.

The registration questionnaire. Everything the consortium does with this application falls under noncommercial use.

- 1 Unpaid, noncommercial usage. This is the branch to take: it needs no billing account and costs nothing.
- 2 The kind of organisation. For the EBD-CSIC and the university partners this is academic or research.
- 3 Confirms the answers. Eligibility is checked against what is declared here, so it is worth reading rather than clicking through.

Noncommercial access covers academic research and teaching, nonprofits, journalism, training, and government bodies doing scholarly research. What it does not cover is fee-for-service work: producing a deliverable that a commercial or government client is paying for. Consortium research, papers, reports to the project and the demonstrations given at project meetings all fall comfortably inside the noncommercial side of that line.

Answer the questionnaire honestly rather than conveniently. Eligibility is checked against what you declare, and a project found to be misregistered loses access rather than being quietly moved.

Finding the project ID

The application asks for the project *ID*, which is not the project *name*. The name is what you typed and can contain capitals and spaces; the ID is lowercase, hyphenated, unique across the whole of Google Cloud, and frequently has digits stuck on the end to make it so.

 **Screenshot pending**

`static/img/manual/project-id.png`

The Cloud console project selector, showing the project ID next to its name.

The project ID is not the project name. The application asks for the ID: lowercase, hyphenated, often with digits appended by Google.

- 1 The display name, which you chose and which the application does not want.
- 2 The ID. This is what goes into the connection form.

Copy it somewhere for step 6. If you lose it, it is on the front page of the Cloud console, in the project selector at the top of every Cloud page, and in the URL of most of them as `?project=...`.

Step 3 — The Earth Engine API

Registering through the page above normally turns the Earth Engine API on for the project by itself. It is worth knowing where the switch is anyway, because a project whose API is off produces a connection that appears to succeed and then fails on the first computation.

 **Screenshot pending**

`static/img/manual/enable-api.png`

The Earth Engine API page in the Cloud console API library, with the Enable button.

The Earth Engine API on the project. The registration page usually turns it on for you; this is where to check when the connection later refuses.

- 1 The project the button applies to. Check it before clicking: the console remembers whichever project you last used, which is not always this one.
- 2 Enables the API. Once on, it reads "Manage" instead.

Open [the Earth Engine API in the API library](https://console.cloud.google.com/apis/library/earthengine.googleapis.com) (<https://console.cloud.google.com/apis/library/earthengine.googleapis.com>), check that the project named at the top of the page is the right one, and enable it if the button offers to. If it already reads **Manage**, the API is on and there is nothing to do.

Step 4 — How much computation you get

Since April 2026 every noncommercial project has a monthly allowance of Earth Engine compute units, measured in EECU-hours, and sits in one of three tiers. New projects start in the first of them.

Tier	Per month	Who it is for
Community	150 EECU-hours	Where every noncommercial project starts. Enough for learning the application and for hydroperiods over single wetlands.
Contributor	1,000 EECU-hours	Still free, but the project must have a billing account attached — attached, not charged. The sensible tier for real work: researchers, graduate students, nonprofits.
Partner	100,000 EECU-hours	By application and manual review, for high-impact climate and biodiversity work. Approval takes several weeks.

 **Screenshot pending**

`static/img/manual/tier.png`

The Earth Engine configuration page showing the project tier and its monthly quota.

The tier decides how much computation the project gets each month. New noncommercial projects land on Community.

- 1 The current tier and its monthly allowance of EECU-hours.
- 2 How much of this month's quota is already spent. It resets on the first of the month.
- 3 Where to move to the Contributor tier, which is still free but wants a billing account attached to the project.

Running out does not lock you out. The project drops into a restricted mode where computations still run but with less parallelism, so everything simply becomes slow. The allowance resets on the first of each month, at midnight Pacific time.

If you intend to compute long Sentinel-2 series over large wetlands, move to the Contributor tier before the meeting rather than during it. Attaching a billing account to a noncommercial project does not cost anything, but it is not something to be discovering while a room waits.

Step 5 — Signing in to the application

The application itself is at <https://restore4life.icts-donana.es> (<https://restore4life.icts-donana.es>). It has no passwords: you give an email address, it sends a link, and following the link signs you in.

Restore4Life Restoring wetlands for a sustainable future

Danube Wetlands HydroApp

Estación Biológica Doñana CSIC

EXCELENCIA SEVERO OCHOA

Home Project site Help

You need an account to work with this application. Authentication uses sign-in links sent by email — there is no password to remember.

Enter your email address and we will send you a sign-in link.

Email address 1

Send sign-in link 2

Hydroperiod analysis powered by [ndvi2gif](#) on Google Earth Engine · [restore4life.eu](#) · [How to cite](#)

The application has no passwords. You give an email address and it sends a link that signs you in.

- 1 Any address you can read. It becomes your account the first time you use it.
- 2 Sends the link. It is good for thirty minutes and for one use.

The address does not have to be the Google one — it identifies you to this application, and has nothing to do with Earth Engine. The first time you use an address it becomes your account.

The link is good for thirty minutes and works once. If it has expired, or if you have already used it, ask for another from the same screen; there is no limit on how many you request.

Step 6 — Connecting the project

Signed in and with no account connected yet, the home page carries a warning saying so, and the link in it leads here. Later on, the same screen is reached from the Earth Engine badge in the bar at the top of every page.

Use Chrome, Edge or Brave for this step. The authorisation flow relies on Google showing you a code to copy, and outside Chromium-based browsers that page can close itself without ever displaying it. This applies only to connecting; once connected, the application works in any browser.

 **Screenshot pending**

`static/img/manual/connect-form.png`

The Earth Engine connection screen, asking for a Google Cloud project ID.

The connection screen, reached from the warning on the home page or from the Earth Engine badge in the bar at the top.

- 1 The project ID from step 3 — the ID, not the display name.
- 2 Opens the Google authorisation step.

Paste the project ID from step 2 and press **Start connection**. The application then shows a link to Google's authorisation page.

Open it, choose the Google account that owns the project, and grant the access it asks for. Google finishes by showing a long authorisation code.

 **Screenshot pending**

`static/img/manual/authorize.png`

The Google authorisation page showing the code to copy back into the application.

Google's own authorisation page. It ends by showing a long code, which is what the application is waiting for.

- 1 The Google account being authorised. Make sure it is the one that owns the project.
- 2 The authorisation code. Copy the whole of it.

Copy the whole code, come back to the application, paste it, and press **Connect**.

📷 Screenshot pending

`static/img/manual/paste-code.png`

The application waiting for the authorisation code to be pasted.

Back in the application, the pasted code is exchanged for a token that is stored encrypted against your account.

- 1 The same link, in case the tab was closed before the code appeared.
- 2 Where the code goes.
- 3 Completes the connection.

The application exchanges that code with Google for a long-lived token, encrypts it, and stores it against your account. The code itself is worthless afterwards and expires within minutes, so there is no harm in a copy of it lingering on your clipboard.

📷 Screenshot pending

`static/img/manual/connected.png`

The top bar of the application showing the green Earth Engine badge with the project name.

Connected. The badge names the project every computation will run on, and clicking it leads back to the screen that can disconnect it.

- 1 The connected project. If this is missing, nothing will compute.

The badge at the top now names your project on a green ground. That is the whole of it — the application is ready, and [the user manual](#) takes over from here.

Disconnecting and reconnecting

The Earth Engine screen of a connected account offers a single **Disconnect** button. It deletes the stored token outright: the application keeps nothing, and the next computation will say there is no account connected.

Disconnecting here does not revoke the permission on Google's side. To withdraw it there as well — worth doing if you are handing the laptop on, or if the account was connected by mistake — go to [your Google account's third-party access page](https://myaccount.google.com/permissions) (<https://myaccount.google.com/permissions>) and remove the entry.

Reconnecting is the same procedure as the first time, and you can connect a different project by disconnecting first: the application holds one project per user, the one every computation runs on.

When it does not work

“That authorization code is invalid or has expired.”

The commonest of them, and almost always one of three things: the code was copied short of its full length, more than a few minutes passed between Google issuing it and you pasting it, or it had already been used. Press **Start connection** again and take a fresh one — codes are single-use by design.

“The Earth Engine connection has expired, start again.”

The connection is spread over two page loads, and the application holds the first half in your session while you are away at Google's page. Signing out, clearing cookies, or leaving it for a very long time drops that half. Nothing is broken; start the connection again.

Google's page never showed a code

Almost certainly a non-Chromium browser — Firefox and Safari are the usual culprits. Repeat the step in Chrome, Edge or Brave.

“Your Earth Engine authorization has expired: connect the account again.”

The stored token has stopped working: the permission was revoked on the Google side, the project was deleted, or the token simply aged out. Disconnect and connect again.

“You have no Earth Engine account connected.”

Exactly what it says, and it appears when a connection was never completed or has just been disconnected. Work through step 6 again.

The connection succeeded but every computation fails

Suspect the project rather than the connection. Check that the ID you gave is the ID and not the display name, that the project is registered ([step 2](#)), and that the Earth Engine API is enabled ([step 3](#)).

Everything has become very slow

The month's allowance is likely spent, and the project is in restricted mode until the first of the month. [Step 4](#) covers the tiers and how to move up one.

Restore4Life — Danube Wetlands HydroApp. Doñana Biological Station (EBD-CSIC), Remote Sensing and GIS Laboratory (LAST-EBD).

Questions and corrections: [the FAQ](#) answers the recurring ones.